No of Pages : 4 **Course Code : 23MX14**

Roll No:
(To be filled in by the candidate)

PSG COLLEGE OF TECHNOLOGY, COIMBATORE - 641 004
SEMESTER EXAMINATIONS, MARCH 2025
MCA Semester : 1
23MX14 DATABASE SYSTEMS

Time : 3 Hours **Maximum Marks :100**

INSTRUCTIONS:
1. Answer **ALL** questions. Each question carries 25 Marks.

1. a) Define composite, multivalued and derived attributes? Give examples for each? (3)

   b) i) With a neat block diagram, outline how data is accessed in a DBMS. ii) Consider you want to build a video site similar to YouTube. The actual video data, the metadata about the video, such as title, the user who uploaded it must be stored and video can be accessed by users simultaneously. Is it better to use a file processing system or a database for storage? What problems do you foresee? Justify. (5+5)

   c) Consider the following information about a university database:

   Professors have an EMP_NO, a name, an age, a rank, and a research specialty. Projects have a project number, a sponsor name (e.g., NSF), a starting date, an ending date, and a budget. Graduate students have an ROLL_NO, a name, an age, and a degree program (e.g., M.S.or Ph.D.).

   Each project is managed by one professor (known as the project's principal investigator).Each project is worked on by one or more professors (known as the project's co-investigators).Professors can manage and/or work on multiple projects.

   Each project is worked on by one or more graduate students (known as the project's research assistants).When graduate students work on a project, a professor must supervise their work on the project. Graduate students can work on multiple projects, in which case they will have a (potentially different) supervisor for each one.

   Departments have a department number, a department name, and a main office. Departments have a professor (known as the HoD who runs the department. Professors work in one or more departments, and for each department that they work in, a time percentage is associated with their job.

   Graduate students have one major department in which they are working on their degree. Each graduate student has another, more senior graduate student (known as a student advisor) who advises him or her on what courses to take.

   Design and draw an ER diagram that captures the information about the university. Identify the entities, relationships, and attributes and also indicate any key and participation constraints applicable. (12)

2. a) Compare Primary key and Unique key in terms of its purpose, NULL value acceptance, Number of keys that can be defined in the table and index created. (3)

b) Define the operations 1. Projection 2. Selection 3. Natural Join in Relational algebra. Demonstrate the operations on the following relations. (6+4)

Table: Student

Rno	S_Name	Faculty_id
1	S1	1
2	S2	8
4	S4	2
5	S5	3
7	S7	3
9	S9	1
10	S10	3

Table: Faculty

Faculty_id	F_Name
1	F1
3	F2
5	F5

Consider the following schema: (12)

c) i) Suppliers (<u>sid</u> : integer, sname : string, address : string)
Parts (<u>pid</u> : integer, pname : string, color : string)
Catalog (<u>sid</u> : integer, pid : integer, cost : real)
The key fields are underlined and domain of each field is listed after the field name

I) Find the name of suppliers who supply some red parts

II) Find the sids of suppliers who supply some red or green parts

III) Find the sids of suppliers who supply some red part or are at 221 packer Ave

IV) Find the sids of suppliers who supply some red part and some green part

V) Find the sids of suppliers who supply every part

VI) Find the sids of suppliers who supply every red part

VII) Find the sids of suppliers who supply every red or green

(OR)

ii) Explain the significant use of relational algebraic expression in RDBMS?

Consider the STUDENT and EMPLOYEE relations, Express the following queries in relational algebra. (6+6)

STUDENT

ROLL	NAME	AGE
1	AMAR	20
2	AKIL	18
3	BALA	19
4	HARSHA	20
5	PRADEEP	21
6	PRADEEP	23

EMPLOYEE

EMPLOYEE_NO	NAME	AGE
E1	ANAND	20
E2	AKASH	23
E3	BALA	25
E4	HARSHA	20
E5	PRANAV	22

1. Find the students with “AGE” 20.
2. Find the names which are in STUDENT relation but not in EMPLOYEE relation.
3. Find the output of the query $\pi_{NAME}(STUDENT) \cap \pi_{NAME}(EMPLOYEE)$.

4. Find the names of the employees whose age is > the maximum age of student.
5. Find the names of student and employees whose name starts with 'A'.

3. a) Differentiate between Single Row Functions and Aggregate Functions with examples? (3)

b) How does SQL perform i) inner join ii) left join iii) right join and iv) full join. For the relation schemes R and S given below demonstrate the join operations. (6+4)

R

A	B	C
u	v	y
v	w	y
y	x	w
y	x	y
u	v	x

S

B	F	C
v	q	Z
x	r	x
v	w	y

c) Consider the following table Employee and SALGRADE, Write the SQL query for the questions below. (12)

TABLE : Employee

ECODE	NAME	DESIG	SGRADE	DOJ	DOB
101	VIKRANT	EXECUTIVE	S03	2003-03-23	1980-01-13
102	RAVI	HEAD-IT	S02	2010-02-12	1987-07-22
103	JOHN CENA	HR	S03	2009-06-24	1983
105	AZHAR	GM	S02	2009-08-11	1984-03-03
108	PRIYAM	CEO	S01	2004-12-29	1982-01-19

TABLE:SALGRADE

SGRADE	SALARY	HRA
S01	56000	18000
S02	32000	12000
S03	24000	8000

I) To display details of all employees in descending order of their DOJ.

II) To display Name and Desig of those employees whose grade is either S02 or S03.

III) To display Name, Desig, SGrade of those employee who joined in the year 2009.

IV) To display all SGrade, Annual_Salary from table SALGRADE where Annual salary= salary *12.

V) To display number of employee working in each Salgrade from the employee table.

VI) To display Name, Desig, Salary, HRA from tables Employee and Salgards where salary is less than 50000.

VII) Select Min(DOJ), MAX(DOB) from employee.

VIII) Select SGrade, Salary+HRA from SalGrade where Sgrade=S02

IX) Select count(distinct sgrade) from employee.

X) Select sum(salary), avg(salary) from salgrade.

4. a) Define partial and transitive functional dependency (FDs) with examples? (3)

b) Discuss the different access control mechanisms provided by DBMS to ensure security? (10)

c) i) Consider the following relation that stores clinical appointments of patients. Using the relation below answer the following questions. (12)

i. Write appropriate column(s) defining the primary key.

ii. State the role of functional dependencies in the process of normalization and define 1NF, 2NF and 3NF.

iii. What is the normal form level of the above relation and what update anomalies are found?

iv. Based on your answer for question (b) fix any anomalies that violate normal form rules and show the dependencies for each of subrelations?

iv. Repeat the steps b and c until you obtain database in 3NF?

Dr_id	Dr_Name	Patient_id	Patient_name	Appointment Date Time	Surgery_no
D101	Sneha	P100	James	12-Aug-23 10.00	S10
D101	Sneha	P105	Jerry	13-Aug-23 12.00	S15
D124	Prem	P108	Iman	12-Sep-23 10.00	S10
D124	Prem	P108	Iman	14-Sep-23 10.00	S10
D132	Robert	P105	Jerry	14-Oct-23 16.30	S15
D132	Rebert	P110	Joseph	15-Oct-23 18.00	S13

What are the objectives of normalization? Comprehend the stages of normalization by mapping the various functional dependencies with respect to each stage?

(OR)

ii) Explain the ACID properties of transaction. What problems can occur if concurrency is not controlled during transaction processing? Enumerate the solutions for implementing concurrency control?

/END/

FD/RL